MIL-P-15737B
27 APRIL 1960
SUPERSEDING
MIL-P-15737A
18 JULY 1951

MILITARY SPECIFICATION

PADDLES, BOAT, LAMINATED WOOD

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers laminated wood paddles for boats and life floats.

1.2 Classification. Paddles shall be of the following sizes as specified (see 6.2):

Length
Inches
48
60
72

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 — Strapping, Flat; Steel.
PPP-B-601 — Boxes, Wood, Cleated-Plywood.
PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 — Boxes, Fiber.
PPP-C-843 — Cushioning Material, Cellulosic.

MILITARY

MIL-P-116 — Preservation, Methods of.
MIL-A-397 — Adhesive, Room-Temperature and Intermediate-Temperature Setting Resin (Phenol, Resorcinol, and Melamine Base).
MIL-T-704 — Treatment and Painting of Material.
MIL-W-6110 — Wood; Determination of Moisture Content of.

STANDARDS

FEDERAL

FED-STD-141 — Paint, Varnish, Lacquer, and Related

FSC 2040

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

Materials; Methods of Inspection, Sampling, and Testing.

FED-STD-595— Colors.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes and Appendix.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification. Laminated paddles furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Description. Paddles shall be of the shape, dimensions, and construction specified herein.

3.3 Materials.

3.3.1 *Wood.* Unless otherwise specifically permitted in 3.3.1.1 and 3.3.1.2 for specific parts, wood used in the construction of laminated wood paddles shall have no splits or openings in the surface of the wood over 1/32 inch deep, and shall contain no wane or bark, rot, dote, incipient decay, wormholes, shake, pith, organic stain caused by various species of fungi, or other strength reducing defects. Natural color of wood, regardless of variability, is not considered a defect. Mineral streak is also not considered a defect. The moisture content of wood used in the laminated paddles shall be not less than 6 nor more than 10 percent.

3.3.1.1 *Handle and shaft.* The handle shall be made of white oak, red oak, birch, hard maple, or of the following species known as "white ash"; white ash, green ash, blue ash, and Biltmore ash. Woods known commercially as "pumpkin ash," "swell-butted," or "brash" material will not be acceptable. The shaft of the handle, and the handle grip shall be made of the same species of wood. Cross grain measured in any direction in the lower 2/3 of the handle and shaft (including the portion attached to the blade) shall not exceed a slope of 1 inch in 15 inches of length. Cross grain measured in any direction in the upper 1/3 portion of the handle and shaft shall not exceed a slope of 1 inch in 12 inches.

3.3.1.2 *Blade.* The blade shall be of yellow poplar, spruce, or basswood, sound, smoothly cut, and well manufactured. Knots less than 3/8 inch in diameter, small burls, and widely scattered pinworm holes and bird pecks will be permitted in the blade. Cross grain not in excess of a slope of 1 inch in 12 inches will also be permitted.

3.3.2 *Adhesives.* The adhesives used for all gluing operations shall be an adhesive conforming to type II, class 2 of Specification MIL-A-397. Only products which have been approved for listing on the applicable Qualified Products List shall be used in the fabrication of paddles.

3.4 Construction. Construction shall be as shown on figures 1 and 2, and as described herein.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T07:54Z
Check the source to verify that this is the current version before use.

3.4.1 *Handle.* The handle grip and shaft may be either of solid or parallel laminated construction. The handle grip shall be glued to the shaft and shall be reinforced by a dowel insert.

3.4.2 *Blade.* The blade shall be of laminated three-ply construction. The two outside plies shall have direction of grain parallel to the length of the paddle and shall be one piece or glued for width. The center ply or cross-banding shall be approximately 1/24 to 1/16 inch in thickness with the grain placed at right angle to the grain of the outside plies, and shall be one piece or glued for width.

3.4.3 *Preparation of blade and shaft for assembly.* The shaft shall be prepared for assembly by drilling a 5/32-inch hole at the extreme depth of the proposed notch, and machining the notch (the hole is to prevent splitting when driving the blade into position). Machining of the notch should preferably be done with a band saw with at least a 1/2-inch blade, operating with a fixture which will insure uniform taper, and straight, smooth surfaces on the sides of the notch. The blade of the paddle shall be grooved on both faces symmetrically about the centerline of the blade preferably with a flat cutting router operated with a fixture which will insure uniform taper and smooth flat surfaces in the bottom of the grooves. The dimensions of the notch in the shaft and of the grooves in the blade shall be adjusted to each other so that when the two parts are assembled the mating faces will fit closely and so that the blade will advance to the full depth of the notch.

3.4.4 *Assembly.* The shaft shall be attached to the blade by gluing. Pressure of 150 plus or minus 25 pounds per square inch (p.s.i.) shall be applied uniformly over the full area of the joint and shall be maintained at the temperature and time required to effect full cure of the adhesive (see 3.3.2).

3.5 Shape and dimensions of paddle. Paddles shall conform to the shape and dimensions shown on figure 2. Except when specifically noted on figure 2, tolerances for the various paddle length, paddle width, and paddle thickness dimensions shall be as specified in table I.

TABLE I. *Dimension tolerances*

Dimension	Maximum average tolerance for a given dimension	Maximum tolerance for individual measurement
	Inch (plus or minus)	*Inch (plus or minus)*
Thickness:		
I, T-1, T-2, T-3, and T-4 (see fig. 2)	1/32	3/32
Other than I, T-1, T-2, T-3, and T-4 (see fig. 2)	1/32	1/16
Width:		
Less than 1 5/8 inch	1/32	1/16
1 5/8 inch to and including 3 inches	1/16	1/8
Over 3 inches	1/8	3/16
Length:[1]		
B, C, and L-1 on figure 2, and total length	1/8	1/4
Radii:		
5/8 inch or less	...	1/32
Greater than 5/8 inch but less than 1 inch	1/16	1/8
Greater than 1 inch but less than 15 inches	1/8	1/4
15 inches or greater	1/2	1

[1] Other length dimensions shall be measured accurate to the nearest 1/32 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

3.6 Straightness. Deviation of the centerline of the paddle from a straight line drawn from the center of the end of the handle to the center of the tip of the blade measured at the mid-blade of the paddle shall be as specified in table II.

TABLE II. *Straightness*

Type of deviation	Average deviation for all paddles in lot	Maximum deviation for an individual paddle
	Inch	*Inch*
Looking flatwise at the paddle:		
48 inch paddle	1/8	3/16
60 inch paddle	1/8	3/16
72 inch paddle	1/8	1/4
Looking edgewise at the blade:		
48 inch paddle	1/8	3/16
60 inch paddle	1/8	1/4
72 inch paddle	3/16	5/16

3.7 Freedom from cupping. Deviation of the centerline of the paddle blade from a straight line extending from edge to edge at the midpoint of the blade shall be no greater than to cause one side of the blade to assume a flat surface.

3.8 Bending strength. Laminated wood paddles shall meet the requirements specified in table III (see 4.7.3).

3.9 Resistance of adhesive bonds to delamination. The maximum delamination in various adhesive lines shall not exceed those specified in table IV (see 4.7.4).

TABLE III. *Bending strength*

Paddle length	Minimum average strength	Minimum strength for any individual paddle
Inch	*Pounds*	*Pounds*
48	255	190
60	230	165
72	200	165

TABLE IV. *Resistance of adhesive bonds to delamination*

Type of adhesive line	Maximum average percent delamination	Maximum percent delamination for any individual paddle
Between laminations of shaft	5	10
Between handle grip and shaft	5	10
Between plies of the blade	5	10
Between blade and the shaft	25	40

3.10 Unpainted paddle surface. Prior to application of coatings, paddles shall be hand or machine dressed, sanded evenly, and smoothly finished with all edges and corners rounded.

3.11 Coatings. Except for the top 4 inches plus or minus ½ inch, paddles shall be coated in accordance with Specification MIL-T-704, type B, except for color. Except for Navy purchases, unless otherwise specified, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

color of the finish shall match color No. 36118 of Standard FED-STD-595. For Navy purchases the color shall match color No. 36076 of Standard FED-STD-595. In addition, the dry finish on the coated paddles shall meet the requirements of 3.11.1, 3.11.2, and 3.11.3.

3.11.1 *Resistance of paint coating to cyclic exposure.* When tested in accordance with the cyclic exposure test specified in 4.7.4, no paint failure shall occur.

3.11.2 *Resistance of the paint coating to accelerated weathering.* No paint failure shall occur as a result of the accelerated weathering test specified in 4.7.6.

3.11.3 *Specular gloss of finish on coated paddles.* When tested in accordance with the specular gloss test specified in 4.7.5, the gloss reading both before and after the accelerated weathering test shall not exceed 8.0.

3.12 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall maintain an effective inspection and quality control system acceptable to the Government to conduct the inspection. It shall be the suppliers responsibility to place in rejection status, paddle lots which have been found defective, until a thorough examination of paddles in the lot have been made, and defective paddles removed.

4.2 Qualification shall be in accordance with 4.2.1 and 4.2.2.

4.2.1 *Facilities.* The prospective supplier is required to submit in detail, information to show adequacy of facilities and personnel for producing, inspecting, and testing of the product. Also shall be included the manufacturer's designation of the approved adhesive, and the names of manufacturers of proposed finishes along with certifications from them of compliance with applicable specification requirements. The information shall be submitted via the cognizant Government inspection office for review and verification.

4.2.2 *Qualification test.* On satisfactorily meeting the facilities requirements of 4.2.1, a qualification test will be authorized at a laboratory satisfactory to the Bureau of Ships to determine conformance with the requirements of this specification. Samples submitted for qualification testing shall be of the quantity and size specified in 4.2.2.1 and 4.2.2.2.

4.2.2.1 *Samples of coating materials.* One quart each of the coatings used, fully identified as to supplier and brand name, shall be submitted for test to determine conformance with the requirements of the applicable governing specification (see 3.11). Tests shall be as specified in 4.7.1.

4.2.2.2 *Specimen paddles.* Eleven finished paddles of 60-inch length shall be submitted for qualification inspection to determine conformance with the requirements of this specification. These shall be manufactured in accordance with regular production procedures.

4.3 Inspection lot. All paddles of the same size, not exceeding 5000, presented at one time for inspection shall be considered a lot for purposes of sampling and inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

4.4 Sampling for examination and tests. Sampling for examination and test shall be in accordance with 4.4.1, 4.4.2, 4.4.3, and 4.4.4. All sampling shall be on an unbiased random basis.

4.4.1 *Sampling for examination of unpainted paddles for surface, separation at adhesive line, and freedom from cupping.* Sampling of paddles shall be in accordance with inspection level II of Standard MIL-STD-105.

4.4.2 *Sampling for examination of paddles for shape, dimensions and straightness.* Sampling of paddles shall be in accordance with the table for small sample inspection shown in Standard MIL-STD-105 at inspection level L-8.

4.4.3 *Sampling for visual examination of paint finish.* Sampling for visual examination of paint finish shall be in accordance with inspection level I of Standard MIL-STD-105.

4.4.4 *Sampling for bending strength test.* Sampling of paddles for the bending test shall be in accordance with the table for small sample inspection shown in Standard MIL-STD-105 at inspection level L-6 when lots contain 301 or more paddles, and at inspection level L-7 when lots contain less than 301 paddles. For individual lot orders of 25 paddles or less the procuring activity may waive the lot acceptance bending strength test (see 6.2).

4.5 Inspection by the contractor.

4.5.1 *Examination of unpainted paddles for quality of surface, separation at adhesive line and freedom from cupping.* Unpainted paddles selected from a lot in accordance with 4.4.1 shall be inspected for quality of surface, separation at adhesive lines, and freedom from cupping. Individual sample paddles with defects shall be rejected. Acceptance or rejection of the lot shall be based on an AQL of 1.5 percent defective for paddles lots with excess cupping and separation at adhesive lines, and an AQL of 2.5 percent defective for quality of surface.

4.5.2 *Examination of paddles for shape, dimensions and straightness.* Sample paddles selected in accordance with 4.4.2 shall be examined for conformance with the shape, dimension and straightness requirements of 3.5 and 3.6, respectively. In measuring dimension tolerance the thickness and width of the handle shaft for each paddle specimen shall be measured at one point. Aso shall be measured one length dimension, one other width dimension, one other thickness dimension, and one radii dimension. The same dimensional measurements shall be made for all paddles in the sample. If as a result of the examination, either the average dimension tolerances requirement of 3.5, or the average straightness deviation requirement of 3.6 is not met, the lot of paddles represented by the sample shall be rejected. For the maximum individual tolerance requirements of 3.5, acceptance or rejection of the lot shall be based on the following defective levels: (a) Thickness and width of handle shaft: AQL of 1.5 percent, (b) Width (except for handle shaft), thickness (except for handle shaft), length and radii: AQL = 4.0 percent. For the maximum individual tolerance requirements of 3.6, acceptance or rejection of the lot shall be based on an AQL of 2.5 percent defective.

4.5.3 *Examination for paint finish.* The paddle manufacturer shall present certifications from the suppliers of the paint coating, that the paint coatings meet applicable specification requirements. It shall be the paddle manufacturer's responsibility to check for conformance with the applicable paint specifications. In addition, the paddle manufacturer shall establish and maintain suitable procedures for his inspection department to effectively examine the paint finishing system, and the painted paddles, to determine conformity with 3.11, excepting 3.11.1, 3.11.2, and 3.11.3. Sampling for examination of paint finish on paddles shall be as speci-

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

fied in 4.4.3. Any paddles in the sample not meeting the finish requirements of 3.11 shall be rejected. Acceptance or rejection of the lot represented by the sample shall be based on an AQL of 4.0 percent defective.

4.5.4 *Bending strength test.* The bending strength tests shall be conducted as specified in 4.7.3, on paddles selected for test by the Government inspector in accordance with 4.4.4, to determine conformance with the requirements specified in 3.8. Tests shall be conducted under surveillance of the Government inspector. If the average value for bending strength of the samples does not meet that specified in 3.8 for a given paddle length, the lot represented by the sample shall be rejected. If the average strength requirement has been met, acceptance or rejection shall then be based on an AQL level of 2.5 percent defective for the minimum strength requirement.

4.6 Inspection by the Government.

4.6.1 *Examination for material, construction, dimension, straightness, freedom from cupping, and surface and paint finish requirements.* The Government inspector shall examine detail inspection records of the contractor in regard to conformance of paddle lots with the material, construction, dimensional, straightness, freedom from cupping, and surface and paint finish requirements of this specification. Spot examination shall also be made to check effectiveness of the firm's inspection. Suspect lots shall be reexamined in accordance with 4.4 and 4.5 of this specification, by the Government inspector and lots rejected or accepted according to findings. Poor or ineffective quality control and inspection by the firm as evidenced by occurrence of defective lots shall result in rejection of paddle lots offered for Government use, until the firm has submitted substantial suitable evidence that paddles with defects have been removed, and defective areas in production, quality control, and examination have been corrected to provide reasonable assurance that future lots submitted for Government use meet specification requirements. Continued failure of lots of paddles offered for Government use to meet specification requirements shall result in withdrawal of qualification approval for the product, and cancellation of the contract.

4.6.2 *Bending strength test.* The Government inspector shall select samples for and observe the bending strength test specified in 4.5.4. Failure of the lot sample to meet that average strength requirement for the given paddle length shall result in rejection of the lot. For the minimum individual bending strength requirement of 3.8 acceptance of the lot shall be based on an AQL of 2.5 percent defective.

4.6.3 *Additional examination and tests.* Where other specifications form a part of this specification, unless otherwise specified herein, sampling, examination, and tests shall be conducted as required in the referenced specifications. Where deemed necessary the inspector may require any of the tests of 4.7 of determine compliance with any of the features of this specification which appear inadequate.

4.7 Tests.

4.7.1 *Tests for suitability of coating materials.* Tests for determining suitability of coating materials shall be conducted as specified in the applicable specification (see 4.2.2.1).

4.7.2 *Moisture content of wood.* The moisture content of the wood shall be determined in accordance with Specification MIL-W-6110. For qualification testing, the oven method shall be used.

4.7.3 *Static bending strength test.*

4.7.3.1 *Apparatus.* The apparatus used for testing wood paddles in static bending shall be such that paddles can be freely supported at a distance 8 inches from each end, and loaded at the distances specified in table V

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

for each paddle length. The apparatus shall be such that the load can be applied in increments of a fairly uniform rate until failure of the paddle occurs. The load indicating device on the apparatus shall be such that it can be easily read in increments of five pounds or less. Between the range of 100 and 350 pounds, any load indicated by the apparatus shall be accurate to a tolerance of plus or minus 2.5 pounds.

TABLE V. *Bending strength test.*

Paddle length	Point of load (measured from end of handle)
Inches	*Inches*
48	22
60	24
72	28

4.7.3.2 *Test procedure.* Each paddle shall be accurately placed in the test apparatus and load applied in increments of a fairly uniform rate until failure occurs. The maximum force indicated by the apparatus during application of load shall be recorded for each paddle. Where more than one paddle specimen represents a lot of a given paddle length, the average load for the paddles in the lot sample shall also be calculated and reported.

4.7.4 *Cyclic exposure test.*

4.7.4.1 *Preparation of test specimens.* Specimens for evaluating various features of the paddle by the cyclic exposure test shall be prepared as specified in 4.7.4.1.1 and 4.7.4.1.2.

4.7.4.1.1 *Preparation of specimens for testing adhesive lines in handle grip.* The specimen used in testing glue bonds in the handle grip shall be prepared by cutting the upper six inches of the handle from the shaft of painted paddles. The paint extending onto this handle section shall be removed by sanding. The sawed surface shall also be sanded smooth.

4.7.4.1.2 *Preparation of specimens for testing adhesive bonds in blade, and adhesive bonds between shaft and blade.* The shaft shall be removed from the paddle by crosscutting at the top of the notch between the shaft and the blade. The blade and remaining attached portion of the shaft shall be bisected by sawing vertically through the thickness of the blade so that the tapered glue lines will be exposed. The sawed surfaces shall be sanded smooth. The edge of each bisected section of the blade shall be sanded smooth to expose glue lines between blade plies.

4.7.4.2 *Test procedure.*

4.7.4.2.1 *Vacuum pressure cycle.* The specimens specified shall be placed in an autoclave, weighted down, and water shall be admitted into the autoclave in quantity sufficient to submerge the specimens completely. During the vacuum pressure cycle water in the autoclave shall be maintained at 65° to 80°F. Thereafter, a vacuum of 20 to 25 inches of mercury shall be drawn and held for 2 hours. The vacuum shall then be broken, and a pressure of 80 plus or minus 5 p.s.i. shall be applied tothe submerged specimens and held for 2 hours. The pressure shall then be released and a vacuum of 20 to 25 inches of mercury shall again be drawn and held for 2 hours while the specimens remain submerged. The vacuum shall be released again and a pressure of 80 plus or minus 5 p.s.i. applied to the submerged specimens and held for 2 hours.

4.7.4.2.2 *Drying period.* The specimens shall be dried for a period of 88 hours (3⅔ days) in air at 80 to 85°F. and 25 to 30 percent relative humidity, and circulating at the rate of 500 plus or minus 50 feet per minute. During the drying period the specimens shall be placed at least 2 inches apart and with the end surfaces parallel to the stream of air.

4.7.4.2.3 *Duration of test.* The cyclical exposure test shall consist of 3 cycles of soaking and drying specified in 4.7.4.2.1 and

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the current version before use.

4.7.4.2.2 comprising a total test period of 12 days.

4.7.4.3 *Measurement of delamination.*[1]

4.7.4.3.1 *Glue lines between blade and shaft.* Delamination of glue joints between blade and shaft shall be measured on the full length of the tapered adhesive lines exposed in bisecting the blade.

4.7.4.3.2 *Glue lines between plies of the blade.* Delamination of glue lines between plies of the blade shall be made on all exposed adhesive lines including those exposed on the curved edges of the blade and those exposed by bisecting the blade.

4.7.4.3.3 *Glue lines at the handle grips.* Delamination of glue joints at the handle grip shall be measured on all exposed adhesive lines. If laminated shafts are used, delamination in the shaft shall be measured and reported separately.

4.7.4.4 *Calculations for percent delamination.* The percent delamination of a particular type of adhesive line in a test specimen shall be calculated as follows:

$$\text{Percent delamination} = \frac{\text{Total delamination measured}}{\text{Total length of exposed adhesive line}} \times 100$$

Average percent delamination for a particular type of adhesive line for all specimens shall be calculated as follows:

$$\text{Average percent delamination} = \frac{\text{Sum of percent delamination for individual specimens}}{\text{Number of specimens}}$$

4.7.4.5 *Observations for paint failure.* Each specimen section of the blade which was subjected to the 12-day cyclic exposure test shall be examined for occurrence of paint failure.

4.7.5 *Specular gloss test.* Before and after coated paddle blades are subjected to the accelerated weathering test specified in 4.7.6, the coating shall be subjected to the specular gloss test specified in method 6101 of Standard FED-STD-141. Prior to conducting the specular gloss test on paddles previously subjected to the accelerated weathering test, the hazy surface deposit shall be removed with a damp cloth, and the surface permitted to dry.

4.7.6 *Accelerated weathering test.* The shaft shall be removed from the paddle by cross-cutting 22½-inches from the tip of the paddle blade. The 22½-inch paddle blade section shall be subjected to the accelerated weathering test specified in method 6151 of Standard FED-STD-141 for a period of 100 hours. At the end of the test period paddles shall be inspected for paint failure.

4.8 **Inspection of preparation for delivery.** Sample paddles and packages shall be selected and inspected in accordance with Specification MIL-P-11[5] to verify conformance to the requirements of section 5 herein.

[1] Delamination is a term used to express separation of wood surfaces at the glue lines. When the separation is in the wood, even though very close to the glue line, it is termed wood failure or checking. Magnification is often necessary to determine whether the failure is in the glue line or wood. A feeler gage, 0.004 inch in thickness is convenient for probing into the glue line to determine if separation actually exists.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* The paddles in quantities of 10 shall be preserved in accordance with method IC-5 of Specification MIL-P-116. The boxes shall be type I, class 2, W5c, style RS of Specification PPP-B-636. The paddles shall be separated from each other with cellulosic cushioning material conforming to Specification PPP-C-843, type II, class C, to prevent damage by abrasion.

5.1.2 *Level C.* The paddles shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* The paddles packaged as specified shall be packed in close-fitting nailed wood boxes conforming to Specification PPP-B-621, class 2, style 2, 2½ or 3, or wood-cleated plywood boxes conforming to Specification PPP-B-601, overseas type, style A or B. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification, except straps shall be flat steel, type I, class B conforming to Specification QQ-S-781. The gross weight of each box shall not exceed 200 pounds.

5.2.2 *Level B.* The boxes packaged as specified (see 5.1.1) shall not require overpacking.

5.2.3 *Level C.* Paddles, packaged as specified shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required, packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Paddles are intended for use with boats and life floats.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Size of paddle required (see 1.2).

(c) Color required (see 3.11).

(d) Whether the lot acceptance bending test for individual orders of 25 paddles or less will be waived (see 4.4.4).

(e) Level of packaging and packing required (see 5.1 and 5.2).

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-15737, whether or not such products have actually been so listed by that date. The attenion of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C., and information pertaining to qualification of products may be obtained from that activity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in/any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:54Z
Check the source to verify that this is the current version before use.

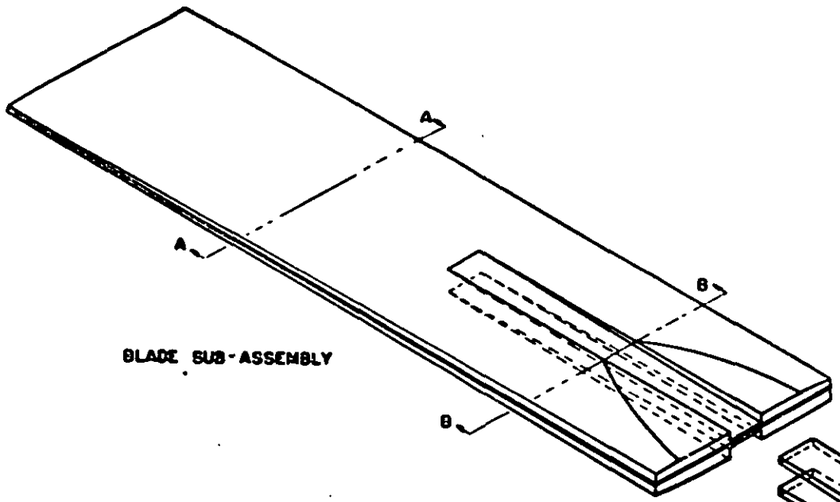
to manufacture use, or sell any patented invention that may in any way be related thereto.

PATENT LICENSE NOTICE. The United States Government has a license to make or have made and to use or have used for Governmental purposes only, paddles covered by U.S. Patent No. 2,399,106.

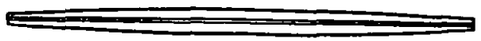
Preparing activity:
Navy—Bureau of Ships
(Project 2040-0036)

Custodians:
Corps of Engineers
Navy—Bureau of Ships

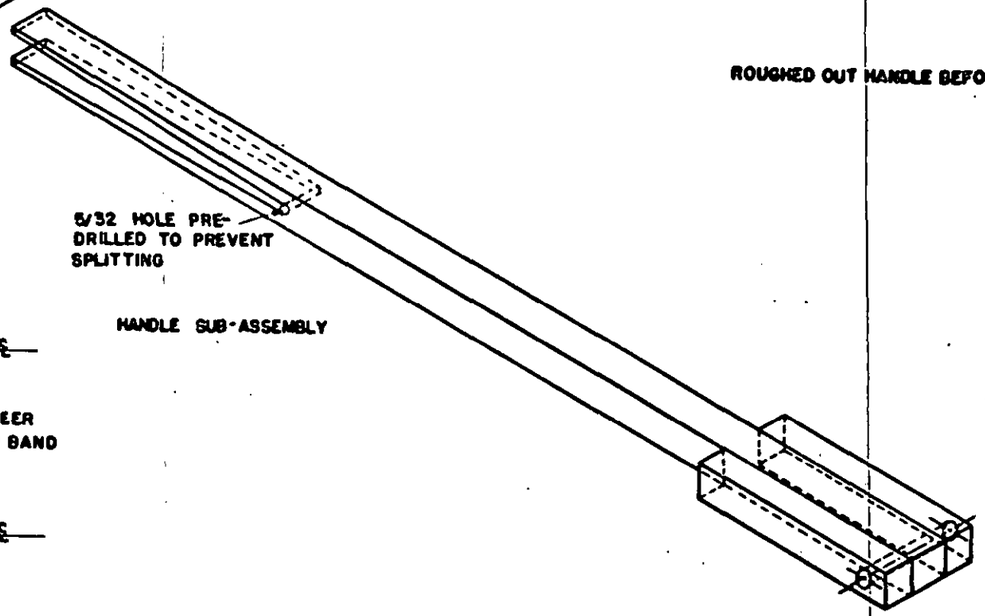
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.



BLADE SUB-ASSEMBLY



TIP END VIEW



HANDLE SUB-ASSEMBLY

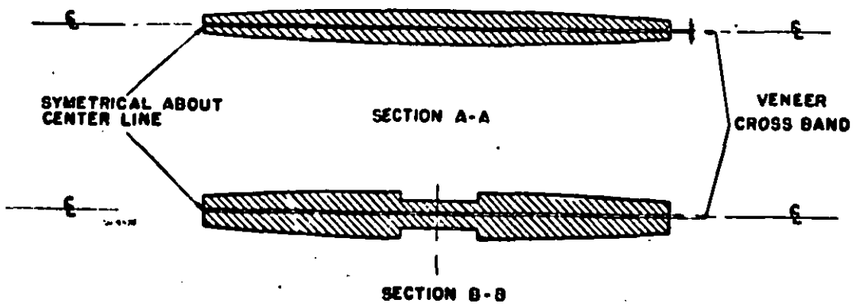
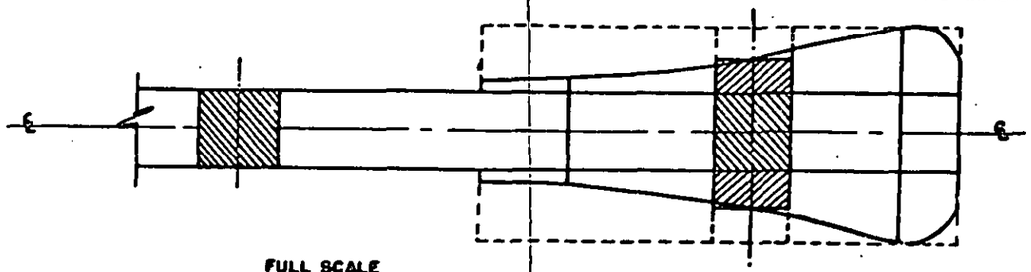
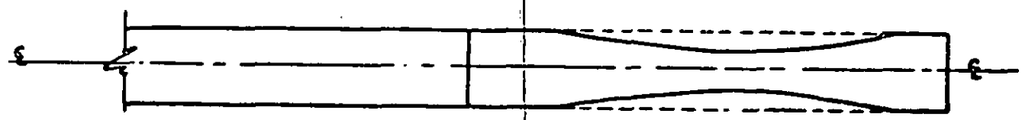


Figure 1.-Basic construction of paddles.

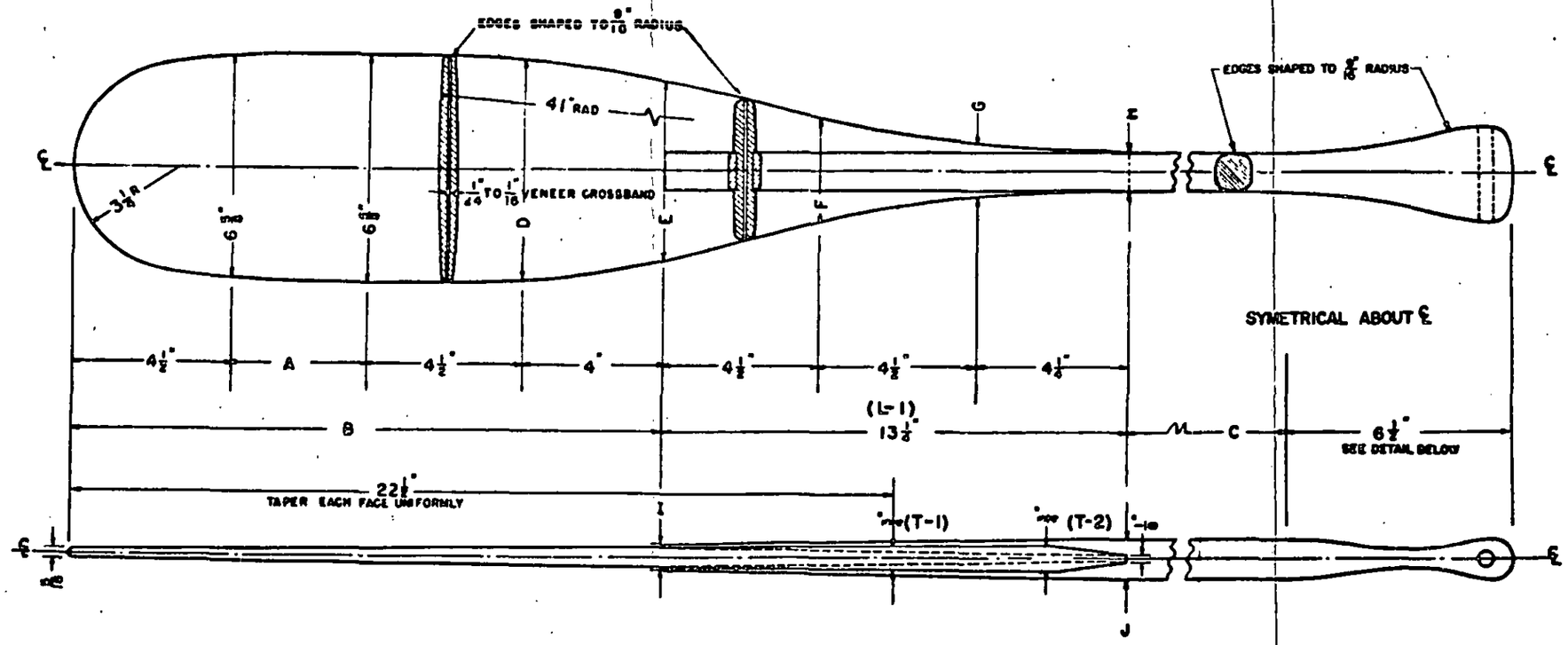


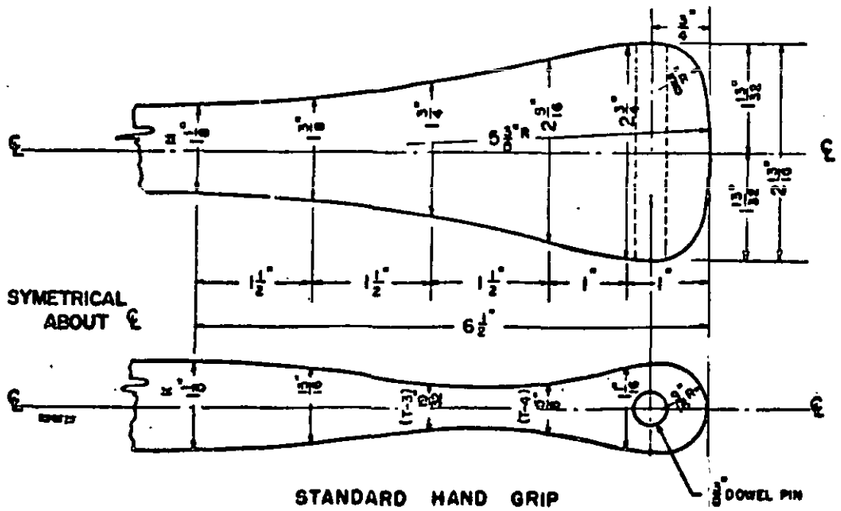
FULL SCALE



ROUGHED OUT HANDLE BEFORE SHAPING

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.





STANDARD HAND GRIP

DIMENSIONS (INCHES)

	A	B	C	D	E	F	G	H	I	J	K	M
SIZE 48 INCH	0	13	15 1/4	6 1/2	5 3/4	3 1/2	1 3/4	1 1/8	11/16	1 1/8	1 1/8	1 1/8
SIZE 60 INCH	4	17	23 1/4	6 1/2	5 3/4	3 1/2	1 3/4	1 1/8	15/16	1 1/8	1 1/8	1 1/8
SIZE 72 INCH	4	17	35 1/4	6 5/8	5 7/8	4 3/8	2 3/8	1 3/16	15/16	1 3/16	1 3/16	1 3/16

NOTE. EASE ALL EDGES BY SANDING

Figure 2.–Shape and dimensions of paddles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:54Z
Check the source to verify that this is the current version before use.